



CORE CS (THEORY)

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❖ Top 50 Computer Networks Interview Questions

1. What is a Computer Network?

A computer network is a collection of interconnected devices that share data and resources. It allows communication between systems using wired or wireless media. Networks improve efficiency and collaboration. Examples include LAN, WAN, and MAN.

2. What are the advantages of Computer Networks?

Computer networks enable resource sharing like files and printers. They support fast communication and data transfer. Networks reduce operational cost. They also allow centralized data management.

3. What is LAN?

LAN stands for Local Area Network. It covers a small geographical area like an office or home. LAN offers high speed and low latency. Ethernet is commonly used in LAN.

4. What is WAN?

WAN stands for Wide Area Network. It covers large geographical areas like countries or continents. WAN uses leased lines or satellite links. The Internet is the best example of WAN.

5. What is MAN?

MAN stands for Metropolitan Area Network. It covers a city or campus area. MAN is larger than LAN but smaller than WAN. It is used by ISPs and universities.

6. What is Network Topology?

Network topology refers to the physical or logical arrangement of devices. Common topologies include bus, star, ring, and mesh. Topology affects performance and reliability. Proper design improves network efficiency.

7. What is Bus Topology?

Bus topology uses a single central cable to connect all devices. Data travels in both directions on the bus. It is simple and low cost. Failure of main cable disrupts the network.

8. What is Star Topology?

Star topology connects all devices to a central hub or switch. It is easy to install and manage. Failure of one device does not affect others. Central device failure can stop the network.

9. What is Ring Topology?

Ring topology connects devices in a circular manner. Data travels in one or both directions. Each device acts as a repeater. Failure of one node can affect the entire network.

10. What is Mesh Topology?

Mesh topology connects devices with multiple paths. It provides high reliability and fault tolerance. Data can take alternate paths. It is expensive and complex to implement.

11. What is Network Architecture?

Network architecture defines the design and structure of a network. It includes hardware, software, and protocols. Common architectures are client-server and peer-to-peer. Proper architecture ensures scalability.

12. What is Client-Server Model?

In client-server model, clients request services and servers provide them. It offers centralized control and security. Servers handle data and resources. Common in enterprise networks.

13. What is Peer-to-Peer Network?

Peer-to-peer network allows devices to act as both client and server. It is simple and low cost. No centralized control exists. Suitable for small networks.

14. What is Protocol?

A protocol is a set of rules for data communication. It defines data format, transmission, and error handling. Protocols ensure reliable communication. Examples include TCP, IP, and HTTP.

15. What is OSI Model?

OSI model is a 7-layer reference model for networking. It standardizes communication functions. Layers include Physical to Application. It helps understand network operations.

16. Name the Layers of OSI Model

The OSI layers are Physical, Data Link, Network, Transport, Session, Presentation, and Application. Each layer performs specific functions. Layers work together to transmit data. It improves interoperability.

17. What is Physical Layer?

Physical layer handles transmission of raw bits. It defines cables, connectors, and voltages. It deals with hardware aspects. It ensures physical connectivity.

18. What is Data Link Layer?

Data link layer ensures error-free data transfer between nodes. It handles framing and MAC addressing. It detects and corrects errors. Ethernet works at this layer.

19. What is Network Layer?

Network layer manages logical addressing and routing. It determines best path for data. IP works at this layer. Routers operate here.

20. What is Transport Layer?

Transport layer ensures end-to-end communication. It provides reliability and flow control. TCP and UDP work at this layer. It manages data segmentation.

21. What is Session Layer?

Session layer manages sessions between applications. It controls dialog and synchronization. It establishes, maintains, and terminates sessions. Useful for long communications.

22. What is Presentation Layer?

Presentation layer formats and encrypts data. It handles compression and translation. It ensures data readability. Also known as syntax layer.

23. What is Application Layer?

Application layer provides network services to users. It supports email, file transfer, and web browsing. It interacts directly with software applications. Examples include HTTP and FTP.

24. What is TCP?

TCP stands for Transmission Control Protocol. It provides reliable, connection-oriented communication. TCP ensures data delivery and error correction. Used in web and email services.

25. What is UDP?

UDP stands for User Datagram Protocol. It provides fast, connectionless communication. UDP does not guarantee delivery. Used in streaming and gaming.

26. Difference between TCP and UDP

TCP is reliable and connection-oriented, while UDP is fast and connectionless. TCP ensures data delivery, UDP does not. TCP is slower than UDP. Choice depends on application needs.

27. What is IP Address?

IP address uniquely identifies a device on a network. It enables communication between systems. IP addresses can be IPv4 or IPv6. It is essential for routing.

28. What is IPv4?

IPv4 is a 32-bit addressing scheme. It provides limited address space. Written in dotted decimal format. Example: 192.168.1.1.

29. What is IPv6?

IPv6 is a 128-bit addressing scheme. It provides a large address space. Written in hexadecimal format. It solves IPv4 exhaustion problem.

30. What is MAC Address?

MAC address is a unique hardware identifier. It is assigned by manufacturer. Works at data link layer. Used for local communication.

31. What is DNS?

DNS stands for Domain Name System. It converts domain names into IP addresses. DNS simplifies internet usage. It acts like a phone directory.

32. What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol. It automatically assigns IP addresses. It reduces manual configuration. Common in LANs.

33. What is HTTP?

HTTP stands for HyperText Transfer Protocol. It is used for web communication. It works at application layer. It transfers web pages.

34. What is HTTPS?

HTTPS is a secure version of HTTP. It uses encryption for data security. HTTPS protects sensitive information. Used in online transactions.

35. What is FTP?

FTP stands for File Transfer Protocol. It transfers files between systems. Works at application layer. Supports upload and download.

36. What is SMTP?

SMTP stands for Simple Mail Transfer Protocol. It sends emails between servers. Works at application layer. Used in email systems.

37. What is POP3?

POP3 retrieves emails from mail server. It downloads emails to client. Emails are usually deleted from server. Simple and widely used.

38. What is IMAP?

IMAP allows accessing emails on server. Emails remain on server. Supports multiple devices. More flexible than POP3.

39. What is Router?

Router forwards data between networks. It works at network layer. It uses IP addresses. Essential for internet connectivity.

40. What is Switch?

Switch connects devices within a LAN. It works at data link layer. It uses MAC addresses. Improves network efficiency.

41. What is Hub?

Hub is a basic networking device. It broadcasts data to all ports. It is less efficient. Mostly replaced by switches.

42. What is Bridge?

Bridge connects two LAN segments. It filters traffic using MAC addresses. It reduces network congestion. Works at data link layer.

43. What is Firewall?

Firewall protects network from unauthorized access. It monitors incoming and outgoing traffic. Acts as security barrier. Prevents cyber attacks.

44. What is Bandwidth?

Bandwidth is the data transfer capacity. Measured in bits per second. Higher bandwidth means faster communication. It affects network performance.

45. What is Latency?

Latency is the delay in data transmission. It affects real-time applications. Lower latency improves performance. Important in gaming and video calls.

46. What is Packet?

Packet is a unit of data sent over network. Data is broken into packets. Each packet contains header and payload. Packets improve efficiency.

47. What is Routing?

Routing determines path for data packets. Routers perform routing. It ensures data reaches destination. Uses routing algorithms.

48. What is Congestion?

Congestion occurs when network traffic exceeds capacity. It causes delays and packet loss. Managed using congestion control. Affects performance.

49. What is Network Security?

Network security protects data and systems. It prevents unauthorized access. Uses firewalls and encryption. Essential for safe communication.

50. What is VPN?

VPN stands for Virtual Private Network. It creates secure connection over public network. VPN encrypts data. Used for remote access.

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